

Two-method Comparision & Plan

Xinyun

Data-based / distribution-based

Main question: Before modeling health outcomes, how should we summarize accelerometer counts?

Me

Outcome-based / association-based

Main question: Given activity summaries or activity patterns, how are they associated with JSW / clinical progression?

Comparison Point	Xinyun's Method	My Method
Core Question	How should accelerometer counts be summarized?	How is activity associated with JSW / clinical progression?
Use of Outcome	Does not use outcome information	Uses outcome information
Method Type	Data-driven / distribution-based	Outcome-oriented / association-based
Research Stage	Pre-modeling activity data summarization	Clinical association analysis after activity features are constructed
Main Contribution	Identifies cohort-specific cutpoints	Evaluates whether activity patterns have clinical relevance
Interpretation of Results	Which cutpoints better preserve the activity distribution	Which activity regions are more strongly associated with the outcome

Result Comparision

Xinyun's result

Counts per minute (0-5780)

Method	t_1	t_2	t_3
Troiano (baseline)	100	200	5,999
L1 optimized	177	1,079	2,755
L2 optimized	595	1980	4487

	Loss1 result	Loss2 result
Troiano thresholds	140547.809745	15664.236299
Loss1 optimized thresholds	25517.513553	3908.838827
Loss2 optimized thresholds	73360.618079	4126.702817

Ideally, L1 threshold should win on Loss 1, L2 threshold should win on Loss 2

L1 appears more preferable under current diagnostics.

1079/5780: approximately 18.67%.

Parameters	J = 60				J = 100				J = 200			
	K=1	K=2	K=3	K=4	K=1	K=2	K=3	K=4	K=1	K=2	K=3	K=4
β_1	0.0036	0.0171	0.0346	0.0344	0.0022	0.0105	0.0183	0.0090	0.0011	0.0052	0.0086	0.0044
β_2	—	-0.0002	0.0072	0.0077	—	-0.0001	0.0042	-0.2306	—	-0.00006	0.0021	-0.5324
β_3	—	—	-0.0023	-0.0366	—	—	-0.0013	0.2204	—	—	-0.0007	0.5254
β_4	—	—	—	0.0012	—	—	—	-0.0009	—	—	—	-0.0004
c_1 (jump)	—	11	3	3	—	18	6	28	—	37	13	58
c_2 (jump)	—	—	23	25	—	—	39	29	—	—	78	59
c_3 (jump)	—	—	—	27	—	—	—	30	—	—	—	60
γ	0	0	0	0	0	0	0	0	0	0	0	0
RSS	796.96	783.78	781.80	781.01	796.89	783.58	781.68	779.52	796.81	783.48	781.63	778.79
BIC	6.31	6.28	17.62	30.20	6.23	6.07	17.51	28.65	6.15	5.97	17.45	27.90

J	c_1 (jump)	c_1/J
60	11	0.183
100	18	0.180
200	37	0.185

• A two-segment model: consistent

• The cutpoint: approximately 18% of the intensity range.

These two results are aligned.

Both methods identify a key boundary around 18% of the activity count range.

Xinyun's data-driven L1 threshold places the sedentary + light boundary at $1079 / 5780 \approx 18.67\%$, while my outcome-based segmentation consistently identifies a cutpoint around 18%. This suggests that the transition from low activity to higher activity may be important both distributionally and clinically.

Sedentary

Light

Moderate

Vigorous

Next Step Arrangement

Xinyun

Next Step:

Apply the optimized activity distribution summaries to the JSW outcome.

Purpose :

Check whether her data-based thresholds are also clinically meaningful.

Me

Next Step:

Identify and adjust for appropriate covariates / confounders

Purpose :

Check whether the activity–JSW association remains stable after adjustment.

Potential convergent finding:

If both methods show that only the sedentary/light activity range is significantly associated with JSW, this would suggest that low-intensity activity contains the most clinically relevant signal in this OAI cohort.

This finding would be especially meaningful because one method is distribution-based, while the other is outcome-based.

Potential Covariate

Basic identifier

Variable	Likely Meaning	Suitable as Covariate?
X	Row index generated during data export	No
ID	Participant ID	No
VISIT	Study visit / follow-up wave	Could
DATE, FVDATE	Assessment date / follow-up visit date	No
TIME	Follow-up time or visit time	Could, for longitudinal or progression models
BARCODE, BARCDBUR, BARCDBUL, BARCDJDR, BARCDJDL	Image or scan identifiers	No

Basic identifier

Variable	Likely Meaning	Suitable as Covariate?
SEX	Sex	Yes
RACE	Race / racial group	Yes
V00COHORT	OAI cohort group	No
AGE_BL	Baseline age	Yes
BMI_BL	Baseline BMI	Yes
AGE	Age at current visit	Yes, but avoid using both AGE and AGE_BL without a clear reason
BMI	BMI at current visit	Yes, if using time-varying BMI
AGE_BL_cat	Baseline age category	Possible, but continuous AGE_BL is usually preferred
BMI_BL_cat	Baseline BMI category	Possible, but continuous BMI_BL is usually preferred

Potential Covariate

Physical activity and general health

Variable	Likely Meaning	Suitable as Covariate?
PASE	Physical Activity Scale for the Elderly	Optional; better for sensitivity analysis
COMORB	Comorbidity score / comorbidity burden	Yes

Knee replacement

Variable	Likely Meaning	Suitable as Covariate?
KRSL	Left knee replacement / surgery status	Usually no; better for exclusion, censoring, or sensitivity analysis
KRSR	Right knee replacement / surgery status	Usually no; better for exclusion, censoring, or sensitivity analysis
ARTL	Left knee arthroplasty-related indicator	Usually no
ARTR	Right knee arthroplasty-related indicator	Usually no

WOMAC symptom

Variable	Likely Meaning	Suitable as Covariate?
WOMTSL	WOMAC total score, left knee	Could, for knee-specific models
WOMTSR	WOMAC total score, right knee	Could, for knee-specific models
WOMTSmax	Maximum WOMAC total score across knees	Yes, especially for person-level models
WOMTSmax_BL	Baseline maximum WOMAC total score	Yes
WOMFNL	WOMAC function score, left knee	Possible
WOMFNR	WOMAC function score, right knee	Possible
WOMPNL	WOMAC pain score, left knee	Possible
WOMPNR	WOMAC pain score, right knee	Possible

Summary

My Next Steps

1

Standard literature review:

Collect commonly adjusted confounders from previous OA / JSW / physical activity studies; Build a defensible adjustment set.

Model recalculation:

Refit your activity–JSW model with these standard confounders; Check whether your low-activity signal remains robust

2

Targeted literature review:

Look for under-addressed confounders, subgroups, or interaction effects; Identify a stronger research gap.

Extended analysis:

Focus on a gap, such as race-by-sex subgroup differences if supported by literature and sample size; Add novelty beyond standard adjusted models.